

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 7 (Ch. 4)

Financial Markets without banks

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies



EUROPEAN CENTRAL BANK | EUROSISTEM

The Governing Council today decided to lower the three key ECB interest rates by 25 basis points. In particular, the decision to lower the deposit facility rate – the rate through which the Governing Council steers the monetary policy stance – is based on its updated assessment of the inflation outlook, the dynamics of underlying inflation and the strength of monetary policy transmission.

The disinflation process is well on track. Inflation has continued to develop broadly in line with the staff projections and is set to return to the Governing Council's 2% medium-term target in the course of this year. Most measures of underlying inflation suggest that inflation will settle at around the target on a sustained basis. Domestic inflation remains high, mostly because wages and prices in certain sectors are still adjusting to the past inflation surge with a substantial delay. But wage growth is moderating as expected, and profits are partially buffering the impact on inflation.

The Governing Council's recent interest rate cuts are gradually making new borrowing less expensive for firms and households. At the same time, financing conditions continue to be tight, also because monetary policy remains restrictive and past interest rate hikes are still transmitting to the stock of credit, with some maturing loans being rolled over at higher rates. The economy is still facing headwinds but rising real incomes and the gradually fading effects of restrictive monetary policy should support a pick-up in demand over time.

The Governing Council is determined to ensure that inflation stabilises sustainably at its 2% medium-term target. It will follow a data-dependent and meeting-by-meeting approach to determining the appropriate monetary policy stance. In particular, the Governing Council's interest rate decisions will be based on its assessment of the inflation outlook in light of the incoming economic and financial data, the dynamics of underlying inflation and the strength of monetary policy transmission. The Governing Council is not pre-committing to a particular rate path.

ECB press release, 30 January 2025

<https://www.ecb.europa.eu/press/pr/date/2025/html/ecb.mp250130~530b29e622.en.html>



EUROPEAN CENTRAL BANK | EUROSISTEM

The Governing Council will stay the course in raising interest rates significantly at a steady pace and in keeping them at levels that are sufficiently restrictive to ensure a timely return of inflation to our two per cent medium-term target.

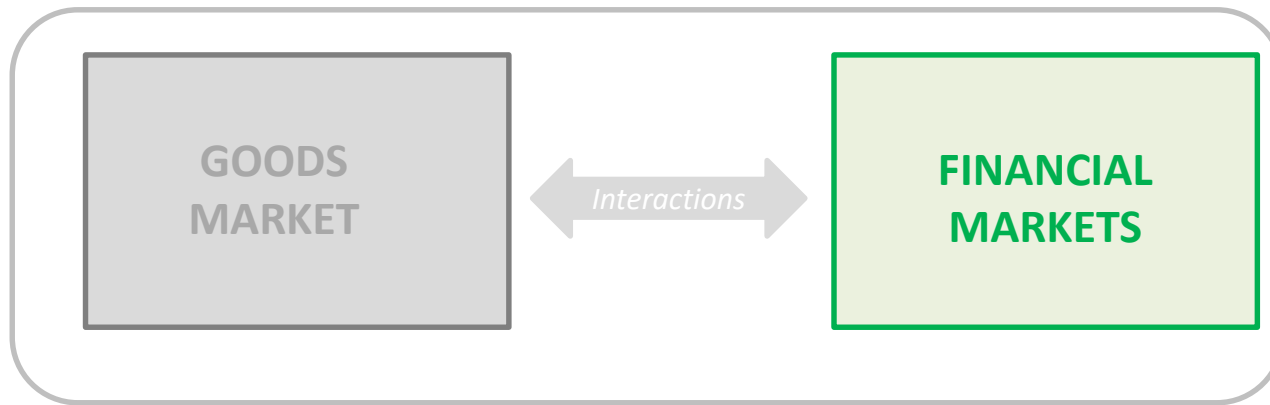
Accordingly, the Governing Council today decided to raise the three key ECB interest rates by 50 basis points and we expect to raise them further. In view of the underlying inflation pressures, we intend to raise interest rates by another 50 basis points at our next monetary policy meeting in March and we will then evaluate the subsequent path of our monetary policy.

Keeping interest rates at restrictive levels will over time reduce inflation by dampening demand and will also guard against the risk of a persistent upward shift in inflation expectations. In any event, our future policy rate decisions will continue to be data-dependent and follow a meeting-by-meeting approach.

ECB press release, 3 February 2023

https://www.ecb.europa.eu/press/press_conference/monetary-policy-statement/2023/html/ecb.is230202~4313651089.en.html

The macro
dynamics of a
country



Keeping in mind our ultimate goal of studying the behavior of the entire economic system, let's first understand how the financial side works

FINANCIAL MARKETS

Investors and financial institutions trade FINANCIAL ASSETS

- Bond
- Stocks
- Money
- More complex assets

As in all markets demand and supply
determine the PRICE
...and, in this case, the RETURN of the asset

WHAT IS MONEY?

Liquid financial assets: directly used in transactions

3 functions:

1. Unit of account for pricing
2. Store of value for saving
3. Medium of exchange for buying things
 - Transactions can be made with
 - Currency
 - Electronic payments thanks to our Bank Deposits

As for all goods there is a market to exchange money: there is a demand for and a supply of money

MONEY MARKET WITHOUT COMMERCIAL BANKS

Money is a good that is traded on a market (money market) based on demand (public uses money to make transactions) and supply (controlled by the “producer” of money: the Central Bank)

MODEL ASSUMPTIONS

- No commercial banks: all money held as currency
- Central Bank is the only financial institution
- Agents own some wealth to allocate in 2 alternative financial assets



MONEY (M)

Perfectly liquid for transactions

No interest

BOND (B): 1 year risk free

Pay an interest, i

Not liquid

Investment problem:
Choosing the optimal allocation btw M and B
implies a trade - off

INTEREST RATE (i)

- Key variable for the money market
- Is the *«price» of money*
 - Opportunity cost of taking M instead of B
 - If I take 1€ as M, I do not invest in B => I loose/pay I
- You are an investor and you buy a 1 year risk free bond
 - $\text{€}P_t$ = today bond price based on D and S
 - $\text{€}R_t = 100\text{€}$ = Face value (amount paid back in 1 year from now) (zero coupon bond)

What is the rate of return paid by the bond?

$$\Delta\% \text{ Asset value} = \underbrace{\frac{\text{€}100 - \text{€}P_t}{\text{€}P_t}}_{\text{\% change btw how much is paid and how much is obtained in 1 year}} \equiv i \quad \leftarrow \text{Nominal interest rate}$$

% change btw how much is paid and
how much is obtained in 1 year

Rearranging we obtain

$$i = \frac{€100}{€P_t} - 1$$

If $€P_t \uparrow \Rightarrow i \downarrow$

If $€P_t \downarrow \Rightarrow i \uparrow$

If $€P_t = 95€$ $i = (100 - 95) / 95 = 0.052$

If $€P_t = 90€$ $i = (100 - 90) / 90 = 0.011$

Given the promise to pay in 1 year, if we pay less today, we obtain more

The price of the bond will be:

$$€P_t = \frac{€100}{1 + i}$$

**Investors'
willingness to pay**

how much we worth the future payments

**Present Discounted Value (PDV)
of the future payment**

today value of the 100€ we will have in 1 year

MONEY DEMAND

- What are the **determinants** of the money demand?

Agents choose how much money to hold based on:

1. LIQUIDITY NEED: transactions in the economy

Proxy - nominal GDP : $\epsilon Y = P_X Y$

If $\epsilon Y \uparrow \Rightarrow \uparrow N/\text{value of transactions} \Rightarrow \text{more money is needed} \Rightarrow M^d \uparrow$

2. MONEY OPPORTUNITY COST wrt BONDS

i = cost of holding money or the «price» of M

$i \uparrow \Rightarrow \text{bonds relatively more convenient than } M \Rightarrow M^d \downarrow$

The money demand is a function of:

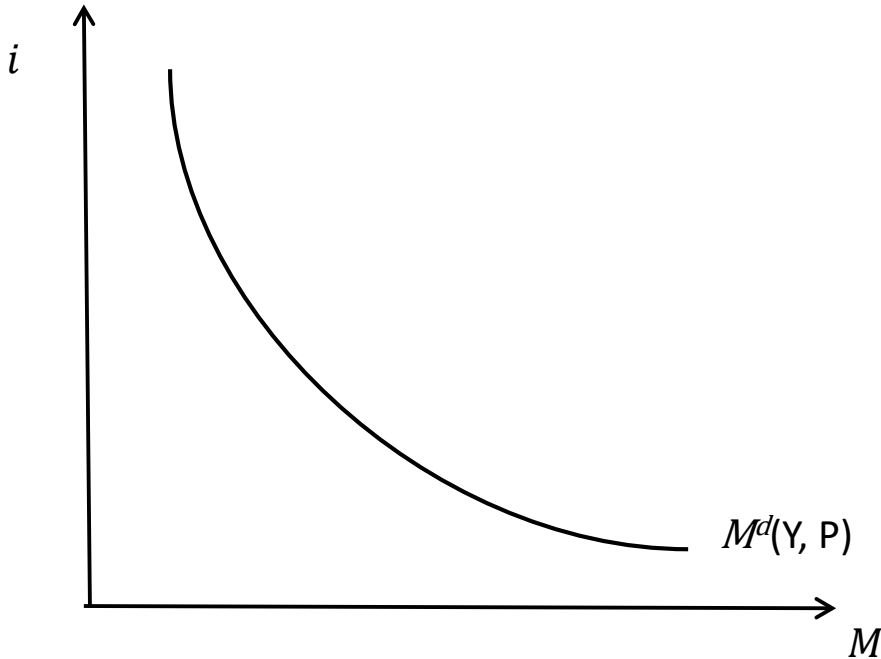
- Level of transactions (Price level and Real GDP)
- Interest rate

$$M^d = \epsilon Y L(i) = P Y L(i)$$

NOMINAL M
demand
function

Liquidity
demand
function

Nominal interest
rate (price of M)

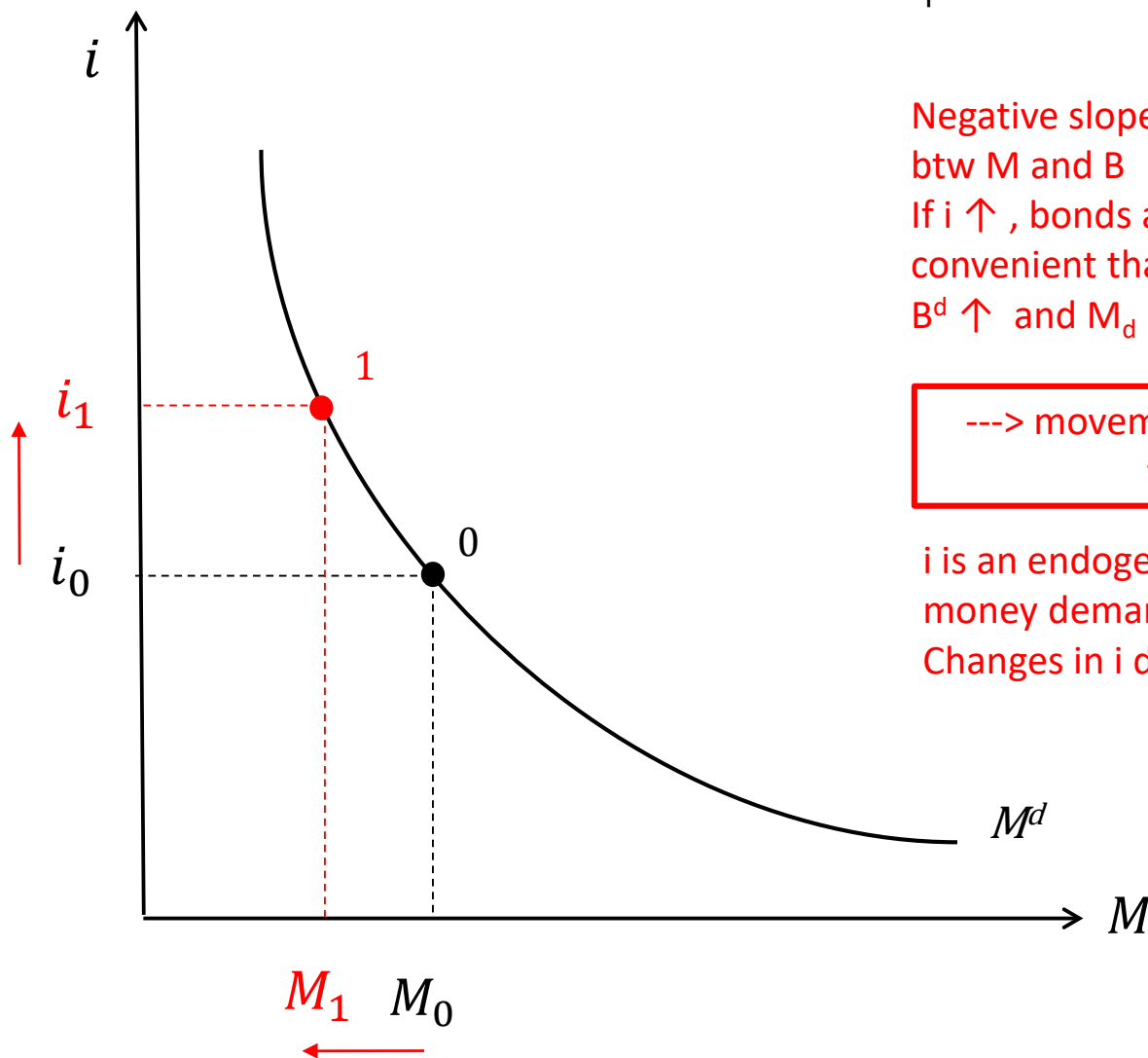


In the (M, i) plane, the M function is plotted for given values of P and Y

SLOPE DETERMINANTS

$$M^D = P \times L(Y, i)$$

+ -



Negative slope since there is a trade off
btw M and B

If $i \uparrow$, bonds are relatively more
convenient than M

$B^d \uparrow$ and $M_d \downarrow$

----> movement ALONG the curve
from 0 to 1

i is an endogenous variable for the
money demand function.

Changes in i do not shift the curve

POSITION DETERMINANTS

$$M^D = P \times L(Y, i)$$

+ -

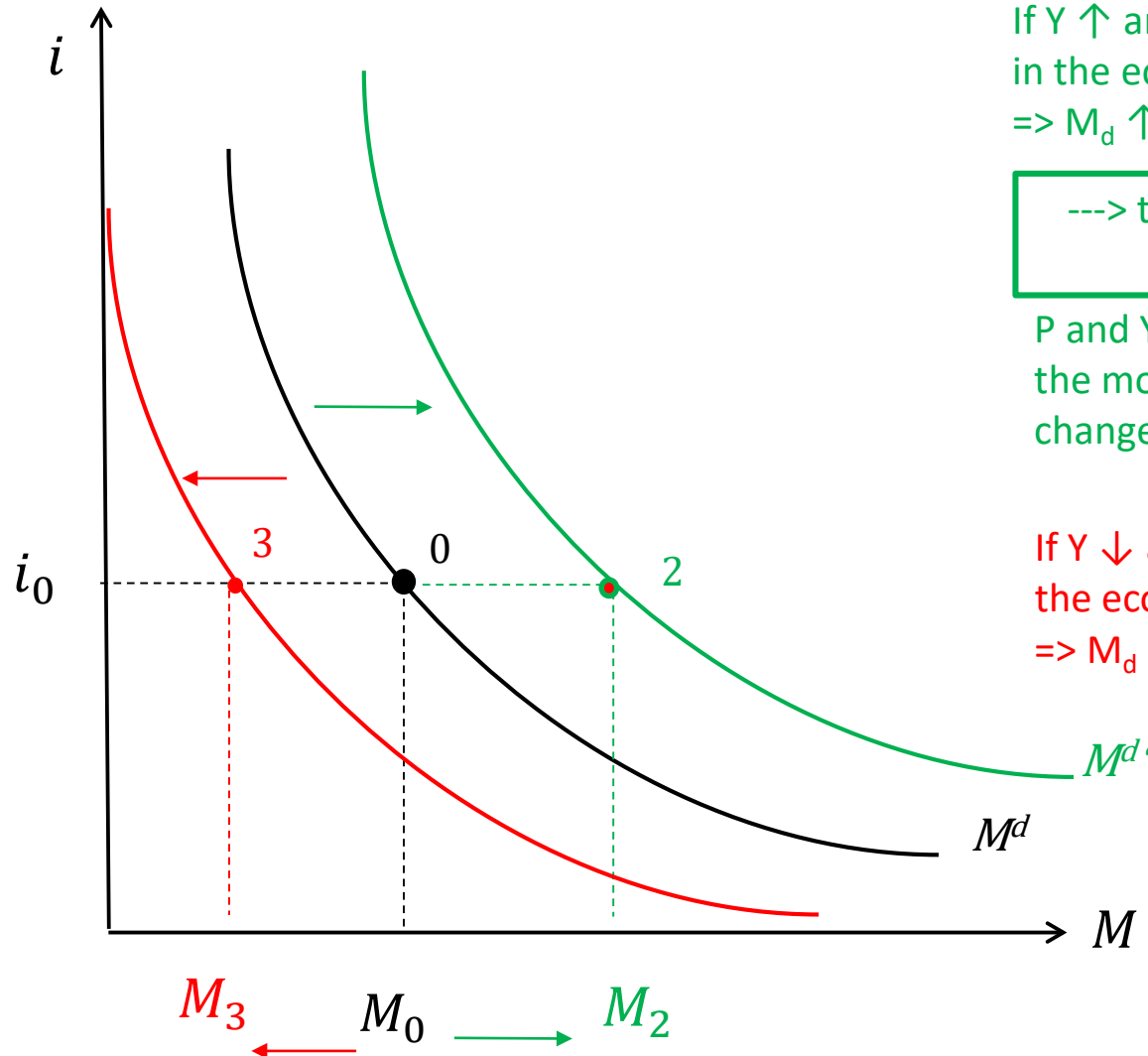
If $Y \uparrow$ and/or $P \uparrow \Rightarrow$ more transactions in the economy and more M is needed
 $\Rightarrow M_d \uparrow$ given i

---> the curve SHIFTS to the right
 passing from 0 to 2

P and Y are exogenous variables for the money demand function. Their change moves the curve

If $Y \downarrow$ and/or $P \downarrow \Rightarrow$ less transactions in the economy and less M is needed
 $\Rightarrow M_d \downarrow$ given i

---> the curve SHIFTS to the left
 passing from 0 to 3



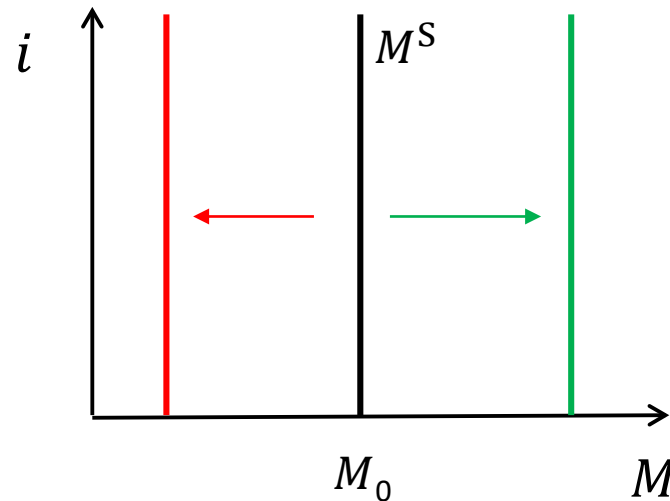
Recap: MONEY DEMAND FUNCTION

$$M^D = P \times L(\underset{+}{Y}, \underset{-}{i})$$

- **Negative slope** : it captures the trade – off btw M and B through i
 - $i \uparrow \Rightarrow M^D \downarrow$ since bonds are more attractive for investors $\Rightarrow$ we move along the money function to the left
 - $i \downarrow \Rightarrow M^D \uparrow$ since bonds are less attractive for investors $\Rightarrow$ we move along the money function to the right
- **Position:**
 - depends on P and Y, taking constant i
 - if P and/or Y $\uparrow$, the M^D shifts to the right since more money is needed for transactions
 - if P and/or Y $\downarrow$, the M^D shifts to the left since less money is needed for transactions

MONEY SUPPLY: DETERMINANTS

- All money held as currency
- Money supply is fully controlled by the Central Bank: chooses the stock of money



- CB uses monetary policy instruments:
 - **Expansionary MP:** CB $\uparrow$ the stock of M supplied
 - **Contractionary MP:** CB $\downarrow$ the stock of M supplied

CENTRAL BANKS AND MONETARY POLICY

How is the monetary policy implemented in practice?

The Eurosystem's instruments



The operational framework of the ECB and euro area national central banks consists of the following set of instruments:

- > Open market operations
- > Standing facilities
- > Minimum reserve requirements for credit institutions
- > Forward guidance

<https://www.ecb.europa.eu/mopo/implement/html/index.en.html>

Open market operations

We use open market operations to steer interest rates, to **manage the amount of liquidity in the financial system** and to signal our monetary policy stance.

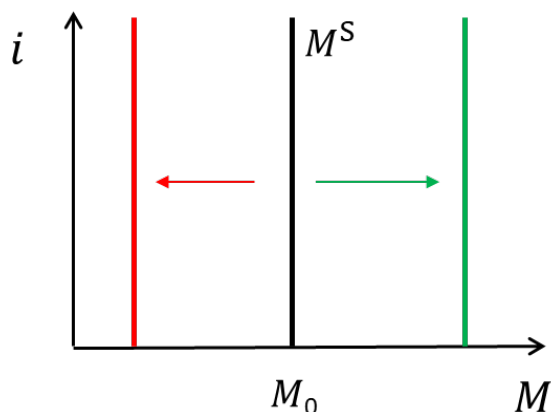
Our regular open market operations consist of one-week liquidity-providing operations in euro (**main refinancing operations, or MROs**) and three-month liquidity-providing operations in euro (**longer-term refinancing operations, or LTROs**).

<https://www.ecb.europa.eu/mopo/implement/omo/html/index.en.html>

Since 2009 the ECB has implemented several measures of unconventional monetary policy

OPEN MARKET OPERATIONS (OMO)

- **What?** Main tool to control liquidity in the economy
- **How?** The CB **sells** (**buys**) bonds (free/low – risk) on the **secondary market** and **is paid** (**pays**) with money



Where already existing
bonds are exchanged
Vs.
Primary market
Where bonds are issued

- **Effect?** They change the CB balance sheet by the same amount

ASSETS	LIABILITIES
Bonds (B)	Money/Currency (M=CU)

- Expansionary MP: The Central Bank **buys bonds** on the secondary market and **pays with money** ⇔ Increase in money supply

$$\Delta M = \Delta B > 0$$

CB buys bonds for 100

ASSETS	LIABILITIES
Bonds (B)+100	Money/Currency +100

- Contractionary MP: The Central Bank **sells bonds** on the secondary market and is **paid with money** ⇔ decrease in money supply

$$\Delta M = \Delta B < 0$$

- CB sells bonds for 100

ASSETS	LIABILITIES
Bonds (B)- 100	Money/Currency - 100

Assets and liabilities change by the same amount

THE MONEY MARKET EQUILIBRIUM

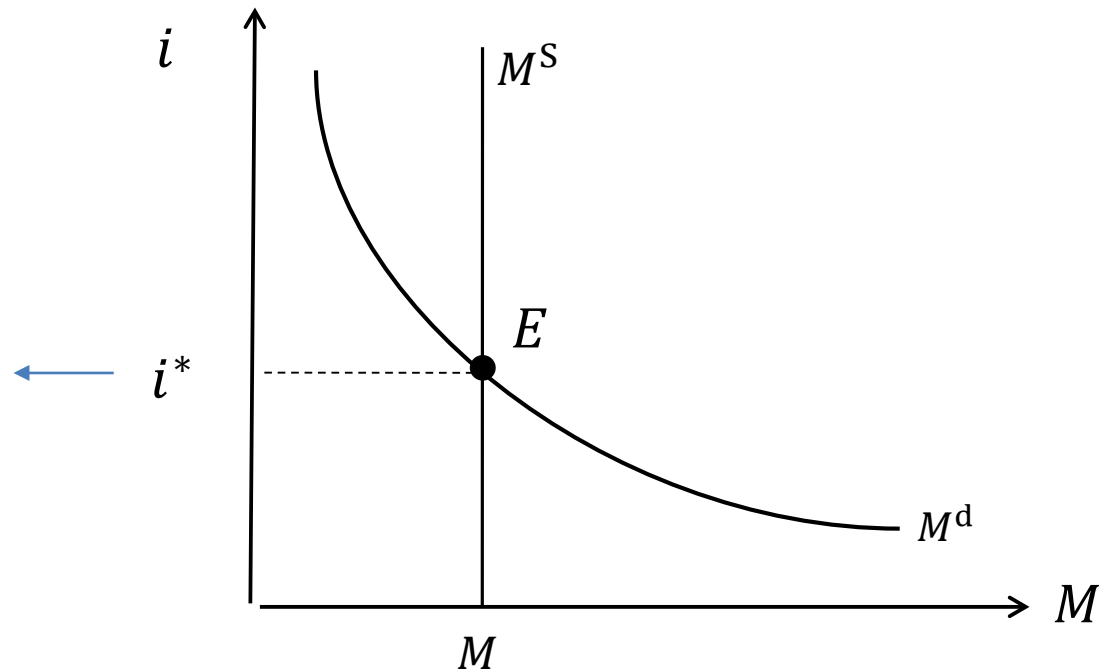
Equilibrium condition on the money market: No excess D nor S

$$M^S = M^d$$

$$M = P Y L(i)$$

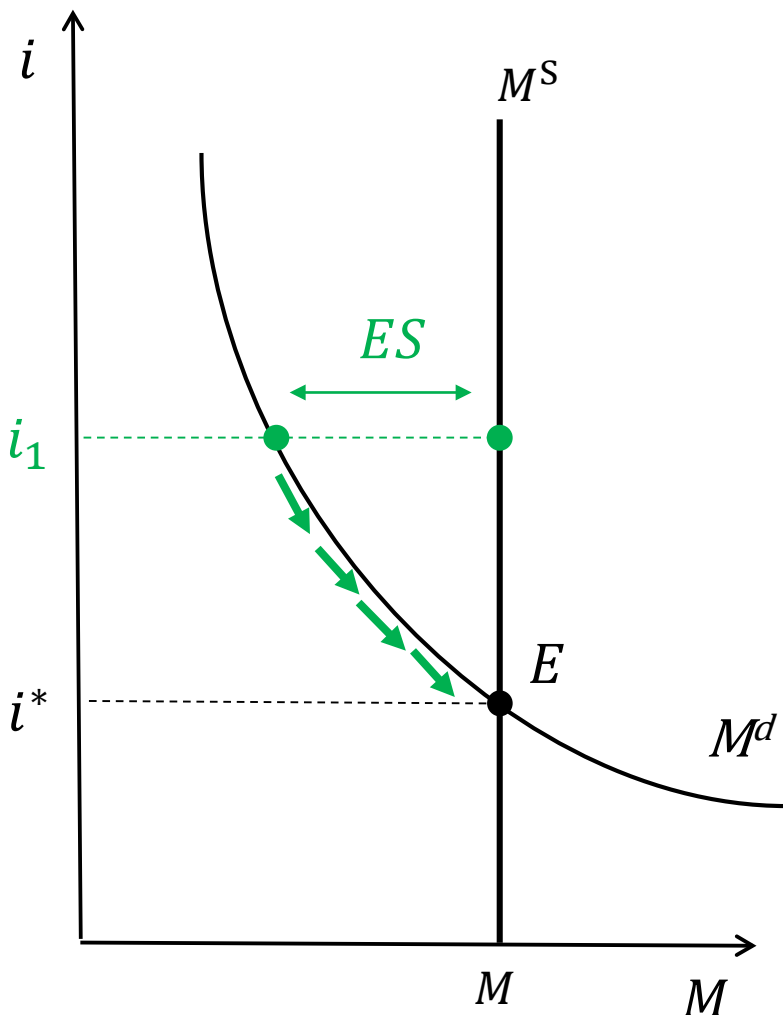
Equilibrium:

The interest rates at which money demand is equal to money supply



WHAT HAPPENS IF THE MARKET IS NOT IN EQUILIBRIUM?

The interest rate has to change in order to restore the equilibrium and to clear the market



$i > i^* \Rightarrow i$ has to decrease

When in the market there is excess supply, the *interest rate* must *decrease*, to *make money more attractive* compared to bonds, increase the demand and close the excess supply

How is it possible? What is the economic mechanism?

i is too high

$\Rightarrow$ Excess supply for M

$\Rightarrow$ Agents use extra money not used for transactions to buy bonds

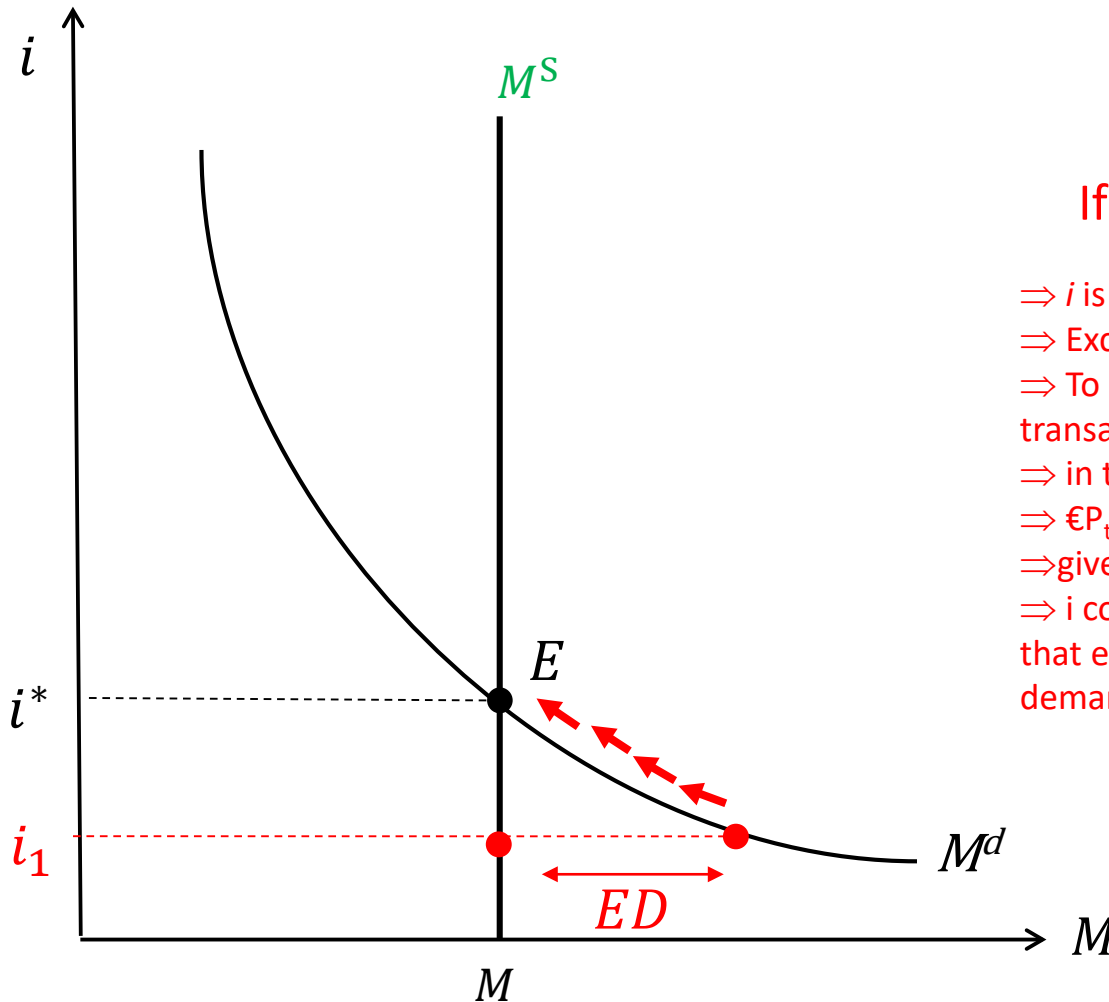
$\Rightarrow$ in the bond market $B^D \uparrow$

$\Rightarrow \text{€}P_t \uparrow$

$\Rightarrow$ given the bonds' face value, $i \downarrow$

$\Rightarrow$ the lower i makes M more attractive for investors that increase their demand up to the point where it is equal to the supply

WHAT HAPPENS IF THE MARKET IS NOT IN EQUILIBRIUM?



If $i < i^*$

- $\Rightarrow i$ is too low
- $\Rightarrow$ Excess demand for M
- $\Rightarrow$ To obtain money necessary for transactions, agents sell bonds
- $\Rightarrow$ in the bonds market $B^s \uparrow$
- $\Rightarrow \text{€}P_t \downarrow$
- $\Rightarrow$ given the bonds face value, $i \uparrow$
- $\Rightarrow i$ continues to increase up to the level that ensures the equilibrium bwt demand and supply for money

EXOGENOUS SHOCK AND COMPARATIVE STATICS

How is the equilibrium modified by the shock?

3-step procedure:

- Which curve is affected?
- Graphical analysis
- Explain the economic intuition

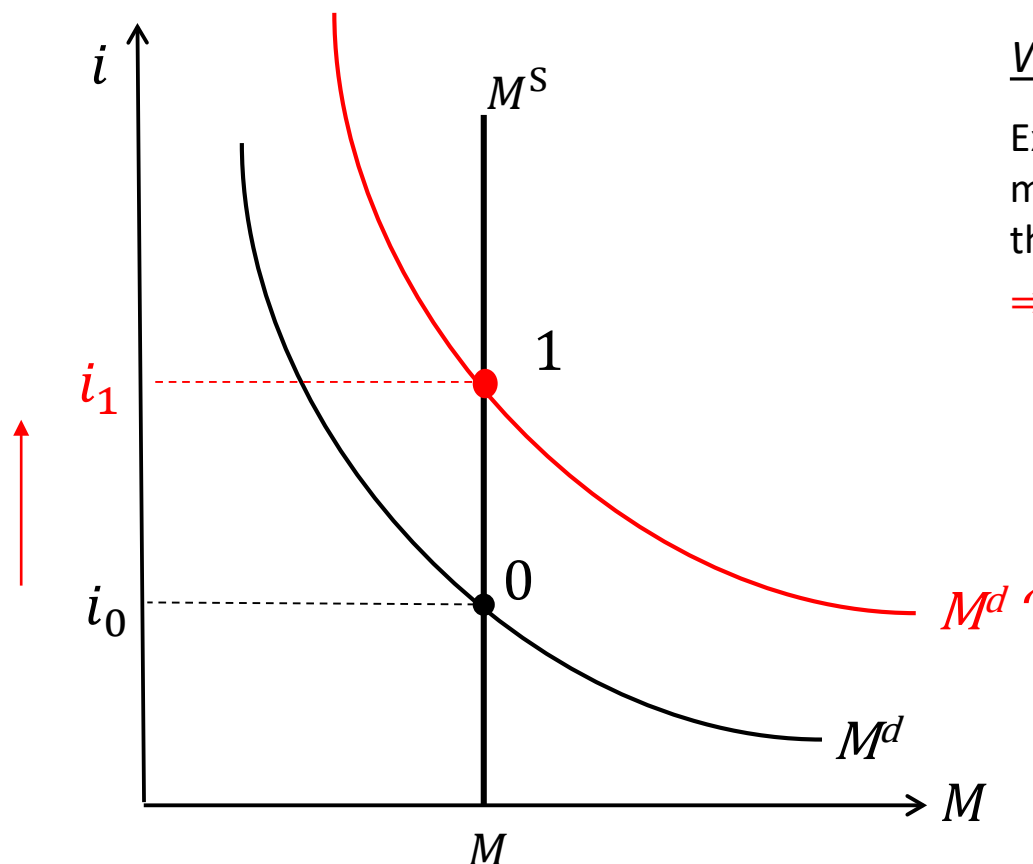
Suppose the economy is in strong recovery phase and there is sudden increase in real GDP

- What we could expect to be the effect on the money market?
- How the equilibrium will be affected?

INCREASE in REAL GDP

Money supply: $= M^s$

Money demand: $M^d = P \cdot Y \cdot L(i)$



Which curve is affected?

Exogenous increase of demand: more income, more transaction, more money demanded by the public

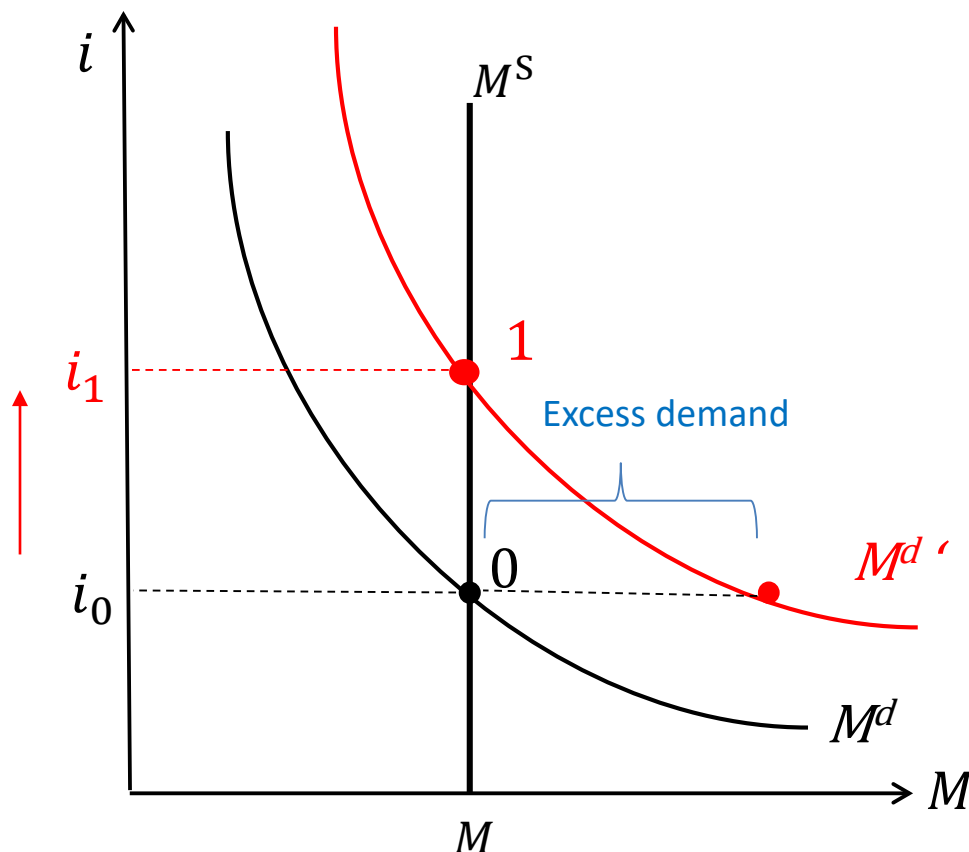
⇒ The money demand curve **shifts right**

What happens to the equilibrium?

The economy moves from 0 to 1: i increases while the nominal money stock is constant

What is the economic mechanism behind the change in the interest rate?

You have to make the reasoning passing from the bond market!



- Exogenous $\uparrow M^d$
- At the initial interest rate, i_0 : $M^d > M^S$
- Excess demand for money
- Agents sell bonds to obtain the extra money they need
 - $\Rightarrow B^S \uparrow$
 - $\Rightarrow \text{€}P_t \downarrow$
 - $\Rightarrow i \uparrow$
- The interest rate must increase, to induce agents to hold a lower amount of money
- As $i \uparrow$, as the demand for money $\downarrow$, closing the excess demand

What happens if $P \downarrow$?

MONETARY POLICY

- Monetary Policy is set by Central Bank
 - EU => BCE
 - US => Fed
 - Japan => Bank of Japan
 - ...
- The CB controls the liquidity level (M^s) in the economy

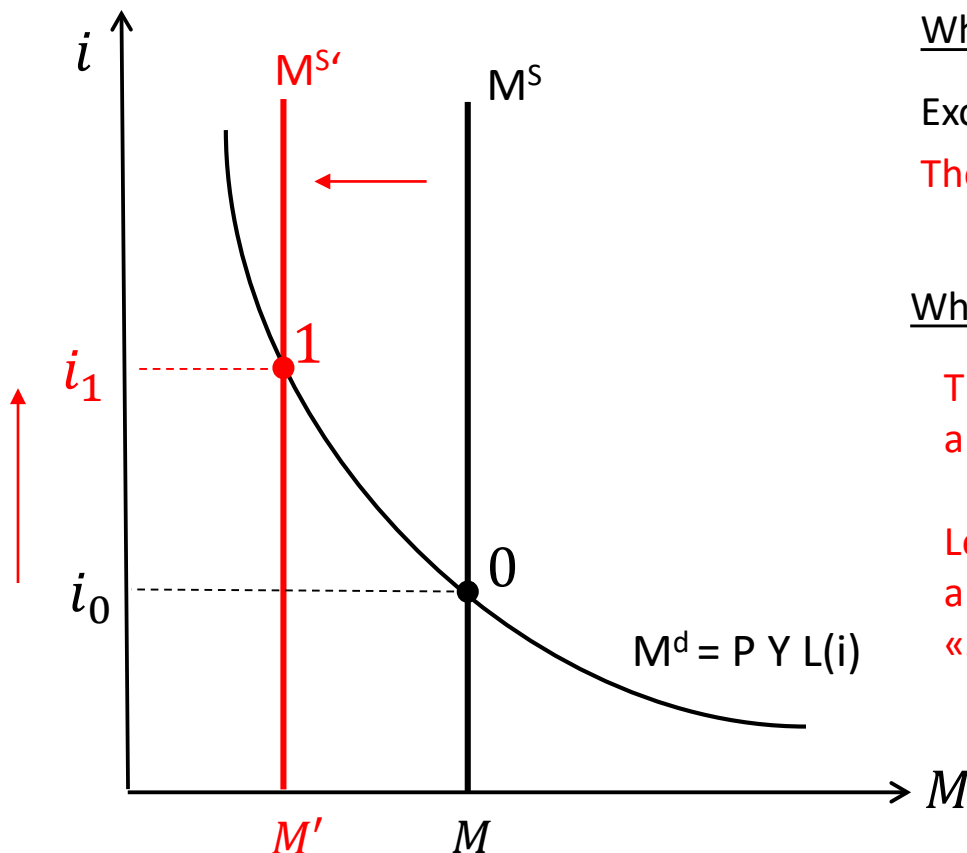
Two key questions:

- 1) What is the impact on the interest rate, i , when the CB injects or reduces liquidity in the economy?
- 2) What is the CB objective?

Suppose the Central Bank decides to reduce the stock of money in the economy

- What we could expect to be the effect on the money market?
- How the equilibrium will be affected?
- Why does the MP affect the interest rate?

CONTRACTIONARY MONETARY POLICY



Which curve is affected?

Exogenous decrease of M^S

The money supply curve **shifts** to the left

What happens to the equilibrium?

The economy moves from 0 to 1: i increases and the nominal money decreases

Less liquidity in the economy translates into a higher price of money: Money becomes «scarce» and its price increases

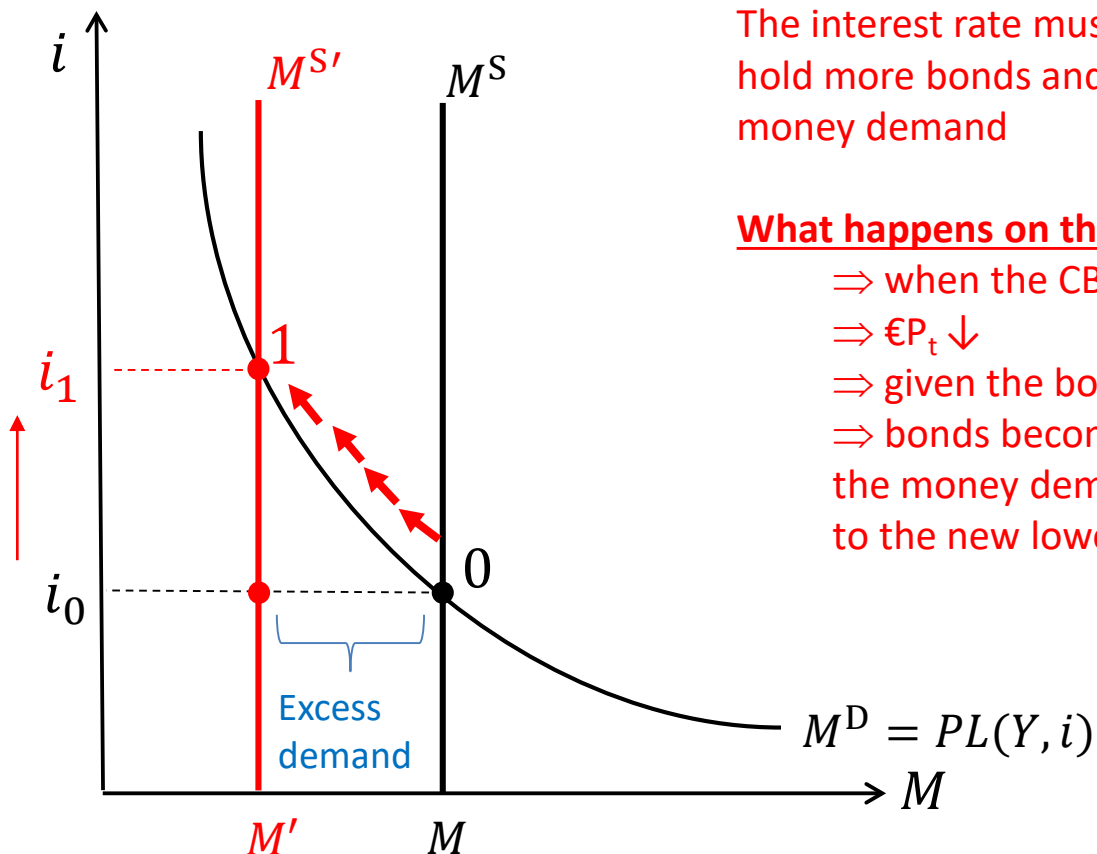
MECHANISM AND ECONOMIC INTUITION

- CB sells bonds and it is paid with money $\Rightarrow$ less liquidity in the economy
- Exogenous decrease of M^S
- At the initial interest rate, i_0 : $M^D > M^S$

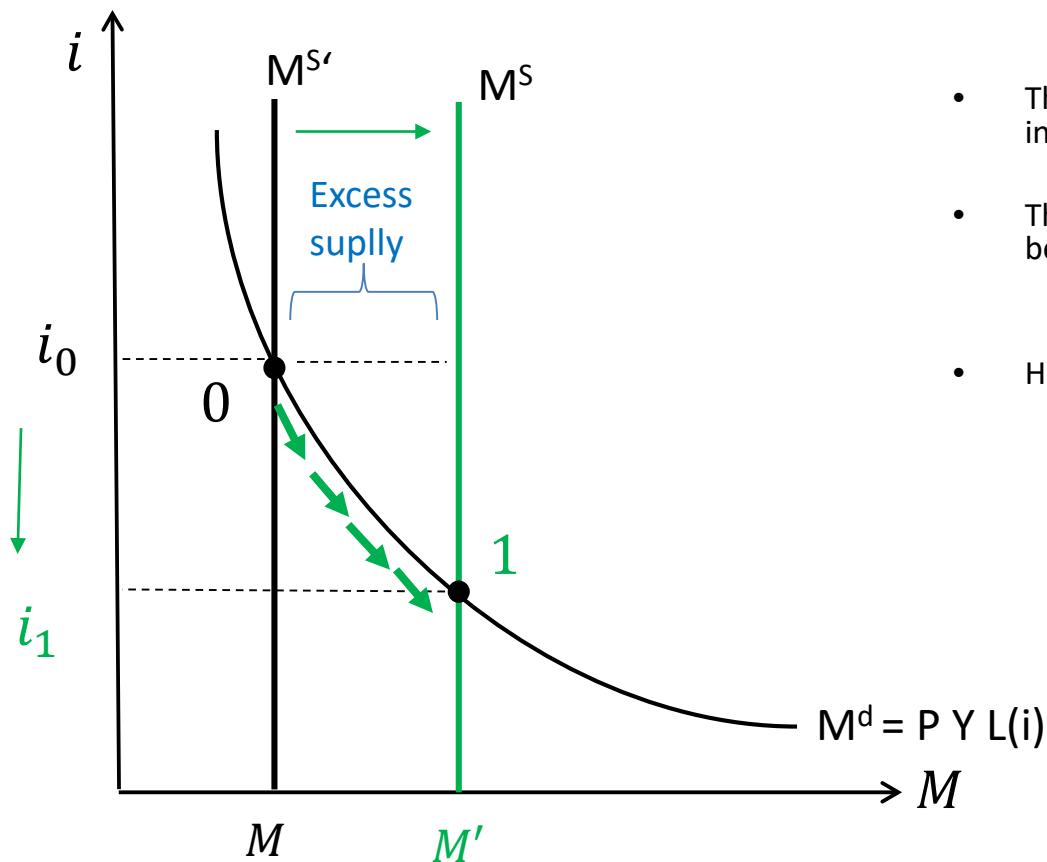
The interest rate must increase, to induce individuals to hold more bonds and less money $\Rightarrow$ to reduce the money demand

What happens on the bond market?

- $\Rightarrow$ when the CB sells bonds the $B^S \uparrow$
- $\Rightarrow \text{€}P_t \downarrow$
- $\Rightarrow$ given the bonds' face value, $i \uparrow$
- $\Rightarrow$ bonds become more attractive for investors and the money demand decreases, up to become equal to the new lower supply



EXPANSIONARY MONETARY POLICY



- CB buys bonds and pays with “new” money
 $\Rightarrow$ Exogenous increase of M^S
 $\Rightarrow$ money supply shifts to the right and an excess supply occurs at the initial equilibrium interest rate
- The interest rate must decrease, to induce individuals to hold less bonds and more money
- The equilibrium moves from 0 to 1: money becomes «abundant» and its price decreases
- How is it possible?
 - On the bond market since the CB buys bonds, the $B^D \uparrow$
 - $\Rightarrow \epsilon P_t \uparrow$
 - $\Rightarrow$ given the bonds' face value, $i \downarrow$
 - $\Rightarrow$ bonds become less attractive for investors and the money demand increases up to become equal to the new higher money supply

MONETARY POLICY: summing up

- Expansionary Monetary Policy
 - The CB **buys** bonds on the secondary market
 - Demand for bonds increases
 - The interest rate declines: given the bonds face value, if their price increase the rate of return decreases
- Restrictive Monetary Policy
 - The CB **sells** bonds on the secondary market
 - Increase in supply of bonds:
 - The interest rate increases: given the bonds face value, if their price decreases the rate of return increases

WHAT DOES THE CB CHOOSE?

Two alternative approaches:

STANDARD CASE:

Endogenous aggregate money

- The Central Bank **sets** the interest policy rate
- The quantity of aggregate money **adjusts** to keep the interest rate constant at the desired level

ALTERNATIVE CASE:

Exogenous aggregate money

- The Central Bank **sets** the quantity of aggregate money
- The **interest rate** is determined by the money demand

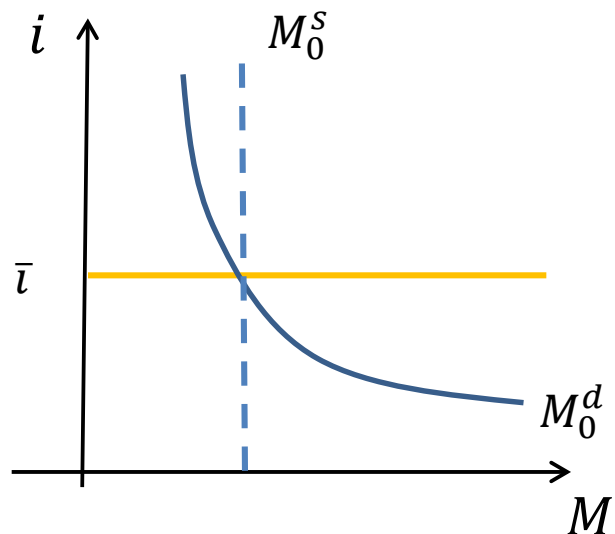


The ECB CASE

What does monetary policy do?

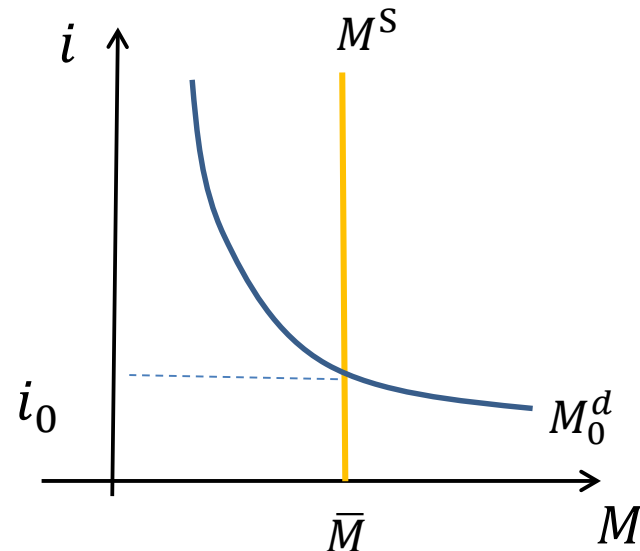
The primary monetary policy instrument is the setting of ECB policy rates, which influence financing conditions and economic developments, thereby contributing to keeping inflation at the ECB's target level.

1. STANDARD CASE:
Endogenous aggregate money



Given the target, once observed the money demand (M_0^d), to reach the target $\bar{i}$, the CB has to adjust money supply at M_0^s

2. ALTERNATIVE CASE:
Exogenous aggregate money



Given the target in terms on money supply, the interest rate is determined by the money demand (M_0^d) and will be i_0

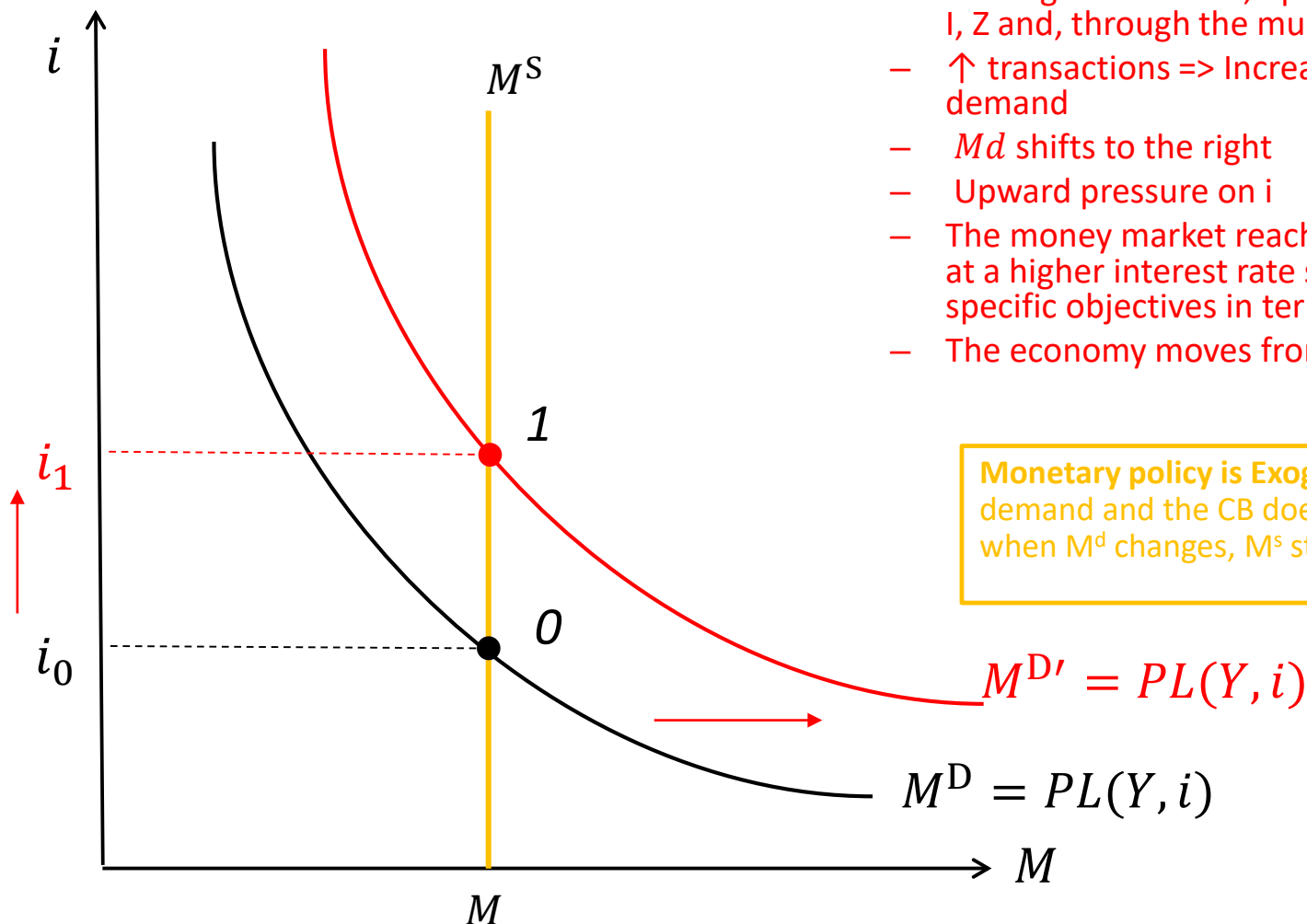
Same shock has different effects depending on the CB behavior.

Suppose that, starting from an initial equilibrium in the money market as described, there is a general wave of optimism due to good news about the global geopolitical situation.

How does the central bank (CB) respond under the two alternative approaches?

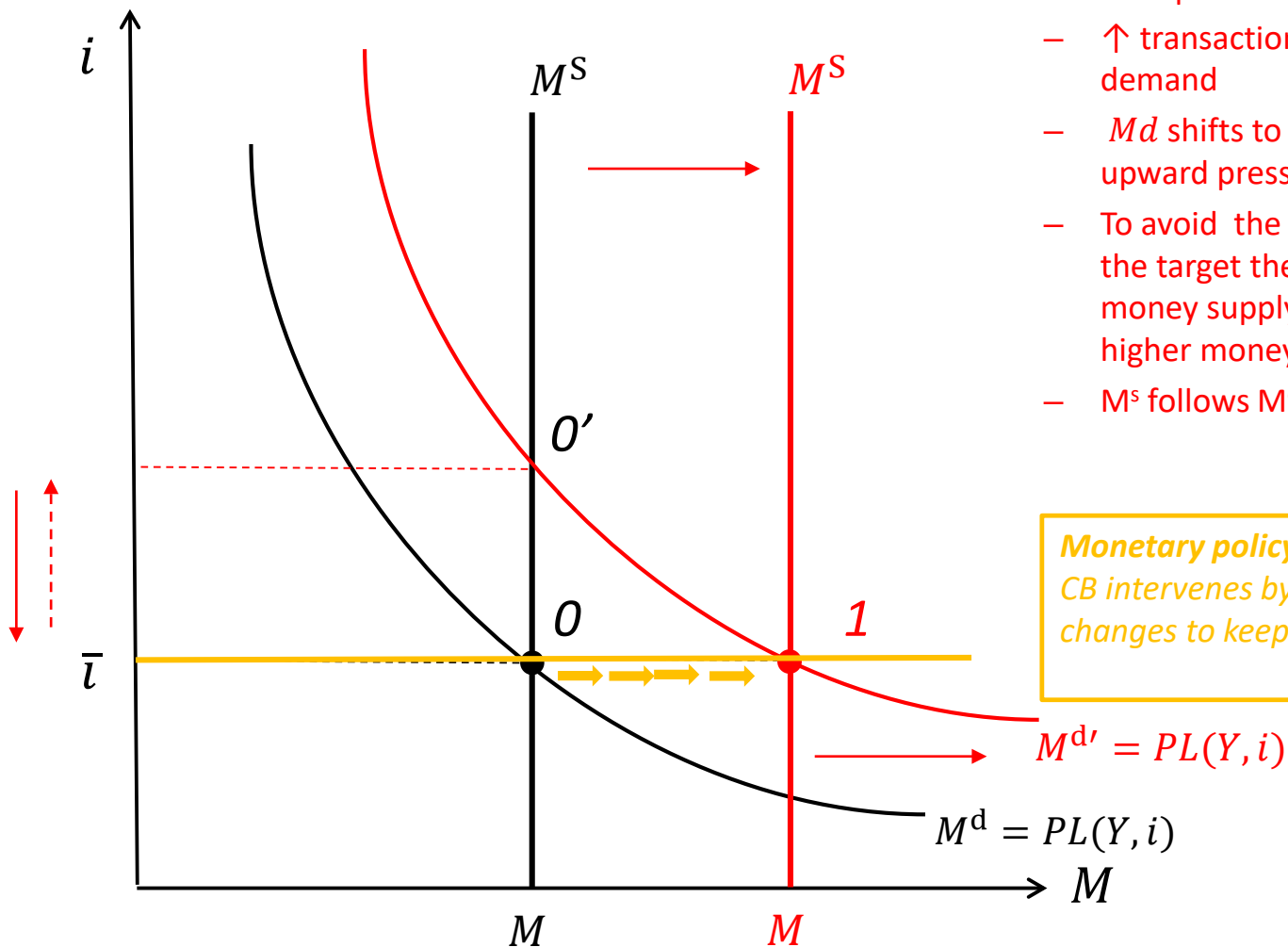
EXOGENOUS MONETARY POLICY

- in the goods market, optimism increases c_0 , I , Z and, through the multiplier, Y
- $\uparrow$ transactions $\Rightarrow$ Increase in money demand
- M^d shifts to the right
- Upward pressure on i
- The money market reaches the equilibrium at a higher interest rate since the CB has no specific objectives in terms of interest rate
- The economy moves from 0 to 1



Monetary policy is Exogenous to the demand and the CB does not intervene: when M^d changes, M^s stays constant

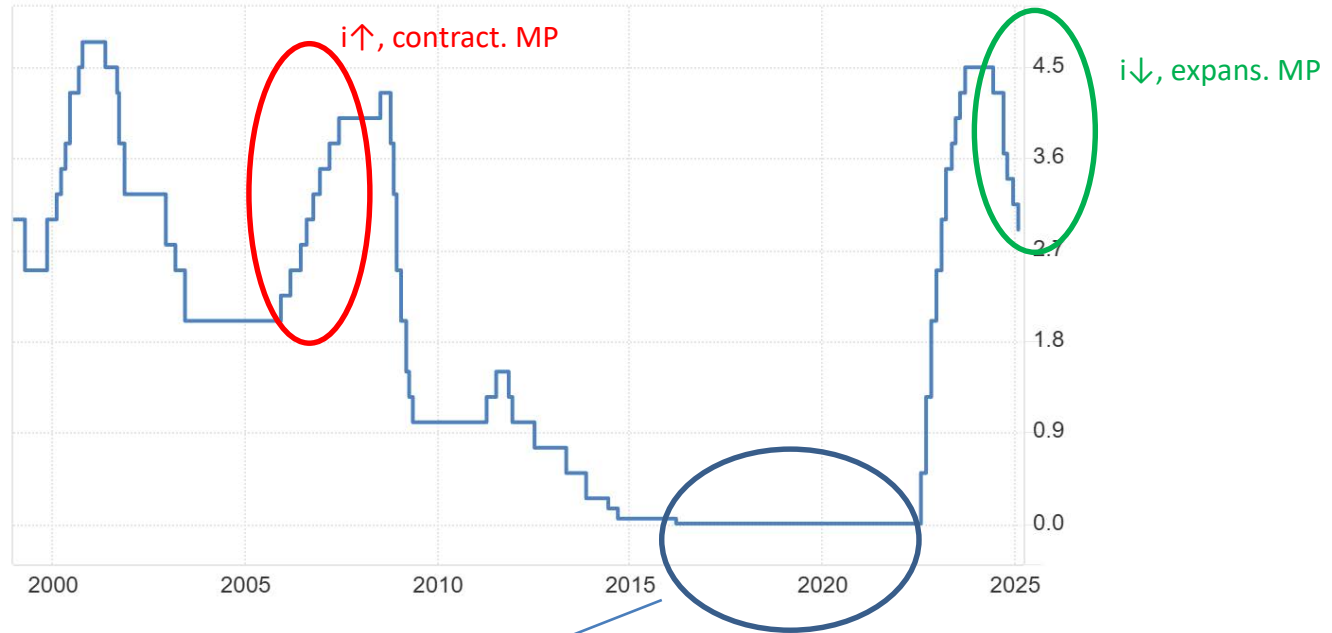
ENDOGENOUS MONETARY POLICY



- in the goods market, optimism increases c_0 , I , Z and through the multiplier Y
- $\uparrow$ transactions $\Rightarrow$ Increase in money demand
- M^d shifts to the right creating upward pressure on i (from 0 to $0'$)
- To avoid the increase in i and to keep the target the CB must increase the money supply accordingly to the higher money demand (from $0'$ to 1)
- M^S follows M^d

Monetary policy is Endogenous and the CB intervenes by changing M^S when M^d changes to keep its interest rate target

EA Interest Rate - percent



Source: tradingeconomics.com | European Central Bank

LIQUIDITY TRAP AND MONETARY POLICY

When $i=0$ the economy is in a liquidity trap and expansionary MP cannot further reduce i

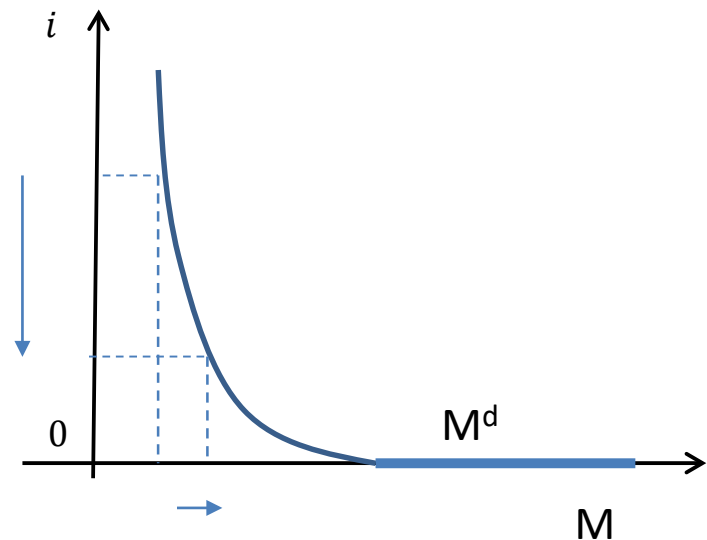
LIQUIDITY TRAP

- The (nominal) interest rate **cannot be negative**

$$i \geq 0$$

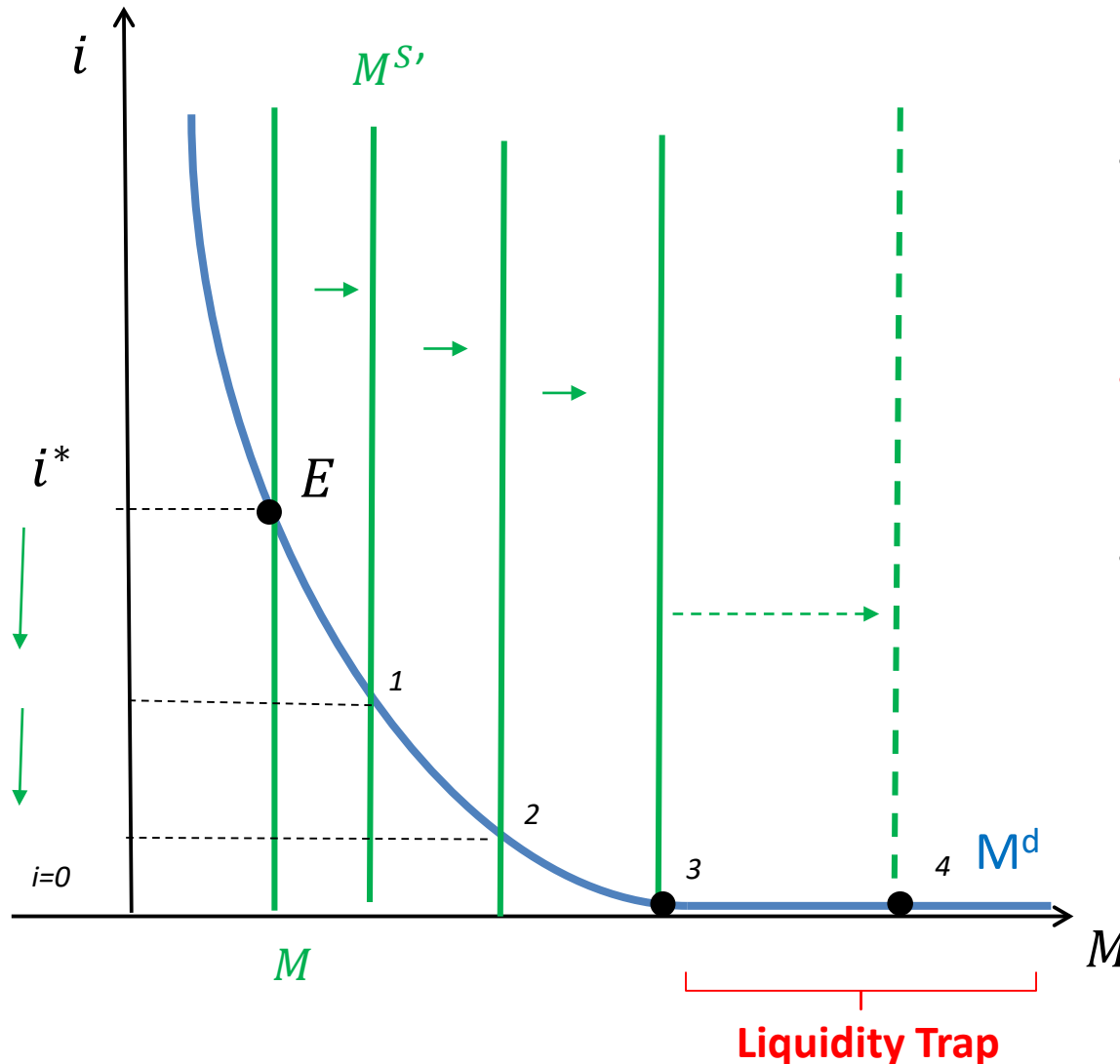
- Why?

- If $i = 0$, bonds are no longer attractive
- Agents are indifferent between bonds and money
- There is no trade off btw bonds and money (M^d is no longer downward sloping)



- When $i=0$
 - The economy is said to be in a Liquidity Trap
 - Conventional MP through OMO is unable to further reduce i

MONEY MARKET: EQUILIBRIUM



- When money demand becomes horizontal expansionary monetary policy has no effect on the interest rate
- More liquidity does not translate into lower interest rate $\Rightarrow$ the economy is in a Liquidity Trap
- The CB cannot reduce the interest rate below zero $\Rightarrow i=0$ is the so called zero lower bound

QUICK CHECK

- When the central bank conducts an open market purchase, what happens on the money market?
 - a. The money supply decreases and the interest rate rises.
 - b. The money supply increases and the interest rate falls.**
 - c. The money demand increases and the interest rate rises.
 - d. Commercial banks reduce their lending.

QUICK CHECK

When the central bank aims to reduce the interest rate, it should

- a. **Buy bonds to increase bond prices.**
- b. Sell bonds to increase bond prices.
- c. Buy bonds to decrease bond prices
- d. Sell bonds to decrease bond prices

QUICK CHECK

Suppose the government increases public expenditures and the central bank has a interest rate policy target. On the money market, the money _____ has to _____ to keep the interest rate constant.

- a. supply, decrease.
- b. supply, increase.**
- c. demand, decrease.
- d. demand, increase.

TAKE AWAY AND KEY CONCEPTS

- Simple case: no banks
 - Price of bonds
 - Money demand:
 - Negative slope
 - Position depends on GPD and prices
 - Money supply: controlled by the CB
 - Contractionary and expansionary MP
 - OMO, bonds price and interest rate
 - Equilibrium: money supply = money demand
 - Comparative statics: monetary policy and shocks to money demand
- Liquidity trap and zero lower bound